



Options 24.x – Sequencing

Sequencing (lead/lag) is managed by using the “setpoint select” function internally to choose between Setpoint 1 and Setpoint 2 via the communications bus between PPC6000's. The “lead” or “master” boiler controls the “lag” or “slave” boiler(s) by switching them from setpoint 2 (“lag stand by” or “banking”) to the setpoint 1 values, and turns on based on Option 24.6 (lag (slave) On Rate) value. The lead boiler will override the modulation rate of the last lag boiler to come on and cause it to modulate in unison with the lead (master) boiler. If both the lead (master) boiler and last lag (slave) boiler remain above the Option 24.6 value another lag (slave) boiler will be started after Option 24.7 (lag (slave) on delay) value has expired. Other lags (slaves) that are on will remain at high fire until the last lag is turned off. At this point, the next lag boiler will begin to modulate with the lead (master) and so on until the lead (master) is carrying the load. The sequence in which boilers are turned on is set in Option 24.1-24.3 via communications addresses of each PPC6000. When a lead (master) boiler is deselected as lead (master) and Option 24.0 has not been changed to 0, the boiler remains as lead (master) until a new lead (master) takes control. Once the new lead (master) takes control, the previous lead (master) may be turned off based on the demand and Option 24.1-24.3 settings.

If the lead (master) boiler is turned off, or fails to come on within three minutes, while operating as the lead, sequencing will be disabled and all lag (slave) boilers will revert to their own PID settings. Should a lag boiler fail to come on within three minutes, or the PPC6000 is faulted, sequencing will immediately call for the next lag in the priority list.

If the lead (master) boiler is switched to manual modulation it will remain as lead bringing on lag (slave) boilers as required. This will allow the operator to “base load” a lead boiler if required.

The PPC6000 can be the “master” (lead) boiler in a system that has PPC5000 or NX3100/4100 series as the communications message are the same. The PPC6000 cannot be controlled by the PPC5000 or NX3100/4100 series controls as a boiler.

Stand by or Banking values

The “lag stand by” or “banking” temperature or pressure is entered as PID 2 or Setpoint 2 values however, options 22.2, 22.3, 22.4 (the P, I and D) values are unnecessary as the burner will start and remain at low fire until the “lag stand by” or “banking” cut out is reached. These values will be entered by;

1. Setting option parameter 22.0 to 1 (Enable Setpoint 2)
2. Setting option parameter 22.1 to the desired setpoint value when in stand by.
3. Setting option parameter 22.5 to 1 (Absolute Value)
4. Setting option parameter 22.6 to the desired cut in value when in stand by.
5. Setting option parameter 22.7 to the desired cut out value when in stand by.

For example: If the normal operating setpoint is 100 PSI, and the desired stand by pressure is 75 PSI. The setup would be:

Option Parameter	Setting
22.0	1
22.1	75
22.5	1
22.6	75 (cut in or minimum pressure)
22.7	80 (cut out or maximum pressure)



STANDBY WATER TEMPERATURE FOR STEAM BOILER

If a steam boiler is being banked based on boiler temperature, a temperature sensor must be fitted into the boiler water jacket. The temperature sensor must be a 4-20mA device. If this device is to be loop-powered by the PPC6000, it should be connected to the 30V output, "sensor supply" on PA16. See option 25.0 – 25.5.

Example:

Fireye Temp. Sensor – TS350	PPC6000	
Sensor Terminal (1) 4-20mA	PPC6000 Terminal	PB8 or PA12
Sensor Terminal (2) Current	PPC6000 Terminal	PA5, PA6, PA7, PA8

The blocks program needs to be 15 or higher.
EK200 shows the current version

Option/Parameters

20.7 = 0 – If using AUX. input 5. Using inputs 1, 2, 3 will allow the boiler water temperature to be displayed on the TSD104. Input 4 will **NOT**.

25.0 = 1 - Analog input 1, (PA5 to PA12 or PB8. Input 2,3 & 4 will also be powered from PA12 or PB8.

25.1 = 32, Zero value of temp. sensor.

25.2 = 350, Span value of temp. sensor.

25.3 = Cut-in temp.

25.4 = Cut-out temp.

No PID involved, lag boiler will turn on and off on low fire.

Relevant Engineers Keys:

EK115, EK120 to EK139 and EK159

NOTE: When using sequencing, setpoint 2 is unavailable for use with any digital input unless the boiler is removed from sequencing completely.

Option 24.0 Sequence Slaves (number of boilers) (0-3) LV1

This option is set to 1, 2 or 3 in each PPC6000 to equal the number of lag "slave" boilers in the system. Once set, option parameters 24.1, 24.2, and 24.3 will become available. If the value is set to 0 the boiler will be taken out of sequencing and operate on its own PID.

NOTE: OPTION PARAMETER 00.6 MUST BE SET TO 0 FOR SEQUENCING TO OPERATE. ON NEWER PPC6000's, Option 0.6 has been moved to the NX610 display. Press and hold the DATA key for 5 seconds. Position the cursor to NORM for sequencing. See Section 4.1 for details.

Option 24.1 -24.3 Priority Number(s) LV1

The order in which boilers sequence on is entered as the communication address(s) of the boilers in **Options 24.1, 24.2 and 24.3**. The first to be turned on will have a priority number one above that of the lead boiler.

For example: If the communications address of the four boilers were 1, 2, 3, and 4 (**set in Option 0.2**), boiler 1 master (lead), with the following entered in **Options 24.1-24.3**



Option parameter 24.x value	Comms. Address	Boiler
24.1	2	2
24.2	3	3
24.3	4	4

The sequencing order would be boiler 2, 3 and 4. If boiler 2 were set to lead via **Option 24.5**, the order would be boiler 3, 4 and 1. Setting boiler 3 to master (lead), the order would be boiler 4, 1 and 2. Other sequencing orders can be achieved by changing the communications address entered in **Options 24.1-24.3**.

Note: Communications addresses should be changed while the boilers are not in sequencing (more than one boiler on) so as not to disrupt operation.

Option 24.4 – Lead OFF Time LV1

DEFAULT 0 (after June 2011)

NOTE: MUST BE SET TO 3 MINUTES (3) OR GREATER FOR PROPER OPERATION.

This option allows the lead boiler to remain in control of all lags up to 999 minutes after the lead is turned off. This may be useful during low water testing or if the burner is recycled for some reason. This is especially useful on low pressure steam boilers.

Option 24.5 – Lead (master) Boiler Select Method LV1

This option parameter selects the lead (master) boiler via the keypad, digital input or communications (Comfire 2). The values are:

0 = Not lead, operates on own PID or as commanded by master (lead)

1 to 3 makes lag boilers modulate to their own setpoint 1 PID when commanded to be ON as follows:

1 = lead (master) via Keypad

NOTE: On newer NX610 Keypads, press and hold the Data key for 5 seconds, then move cursor to for right. The word “lead” in lower case indicates the keypad has this boiler set to lag. Pressing the Com key will cause the “lead” to switch to upper case “LEAD”. This indicates the boiler is now Lead via Keypad. Pressing the Data key return you to the normal screen.

2 = lead (master) via digital input. **(Digital Input Selected via Option 20.8)**

3 = lead (master) via communications such as Comfire 2.

4 - Not a lead boiler (lag only) - Same as setting "0"

5 to 7 makes lag boilers modulate in unison with the lead when commanded to be ON as follows

5 = lead (master) via Keypad as above.

6 = lead (master) via digital input. **(Digital Input Selected via Option 20.8)**

7 = lead (master) via communications such as Comfire 2.

Note: When a lead (master) boiler is deselected as lead (master) and Option 24.0 has not been changed to 0, the boiler remains as lead (master) until a new lead (master) takes control. Once the new lead (master) takes control, the previous lead (master) may be turned off based on the demand and Option 24.1-24.3 settings.



Option 24.6 – lag (slave) On Rate (0-100%) LV1

This value determines the firing rate of the lead (master) boiler at which the first and all subsequent slave(s) (lag) boilers come on. The slave(s) (lag) boiler(s) will be delayed in coming on by the time set in **Option parameter 24.7**. At this point, the lead (master) and the last "lag on" will begin to modulate in unison. If the lead (master) and modulate above this point, the next, as determined by priority number (**Option 24.1, 24.2, 24.3**), will be turned on. When more than one boiler is on, the master and last slave will modulate in unison while the other slaves modulate on their own PID,

Option 24.7 – lag (slave) On Delay (0-999 minutes) LV1

This option sets the time delay between the modulation rate of the lead (master) boiler (and last boiler on if any) exceeding the value set in **Option 24.6** and the next being turned on.

This time may also include additional time as set in **Option 25.6**.

Option 24.8 – lag (slave) Off Rate (0-100%) LV3

When the lead (master) boiler and last boiler modulate below this modulation rate and exceed the value set in **Option 24.9**, the last turned on will be turned off. At this point, the lead (master) and the previous (if any) lag turned on will begin to modulate in unison until they together are below this setting. This will continue until only the lead (master) boiler remains on.

Option 24.9 – lag (slave) Off Delay (0-999 minutes) LV3

This option sets the time delay between the modulation rate is at or below the value set in **Option 24.8** and the last turned on being turned off. This also sets the time delay between successive slaves (lag) being turned off should the modulation rate stay below the setting in **Option 24.8**.

Option 25.0 – Banking based on temperature / water temperature standby for steam boiler LV3

This option sets which input will be used and the 4-20mA current input from the temperature sensor must be connected into one of the 5 possible current inputs:

- 0 = lag banking on temperature is not enabled.
- 1 = uses analog input 1 (PB8 to PA5 input)
- 2 = uses analog input 2 (PB8 to PA6 input)
- 3 = uses analog input 3 (PB8 to PA7 input)
- 4 = uses analog input 4 (PB8 to PA8 input)
- 5 = uses analog input 5 (PA12 supply to PA13 input)

NOTE - If a loop-powered sensor is used, the "+" to the sensor is taken from PB8/PA12 ("sensor supply", JP3 must be set to the "30V" position) and the "-" from the sensor is connected to PA5, PA6, PA7, PA8 or PA13 depending on which input number is to be used. If the temperature sensor gives a current feedback that is out of range, i.e. Less than 3.5mA or greater than 21mA, lag banking on temperature will be cancelled and the sequence lag boiler will bank based on steam pressure.



The following option parameters are only visible if option 25.0 is non-zero:

Option 25.1 – Water temperature zero value LV3

Sets the zero value for the 4-20mA input (usually 0 for a 0-xxx degree temperature sensor).

Option 25.2 – Water temperature span value LV3

Sets the span value for the 4-20mA input (i.e. the sensor range xxx).

Option 25.3 – Water temperature cut-in LV1

Sets the low banking cut-in temperature.

Option 25.4 – Water temperature cut-out LV1

Sets the high banking cut-out temperature.

Option 25.5 – Temperature based warming limit LV1

This works independently and in addition to the normal warming limit. If a non-zero value is entered in 25.5, the burner will also hold at low fire, if the temperature drops below this limit.

Option 25.6 – Lag wait time LV1

This option allows for additional time to be added before a second or third lag is started. This allows the previous lag time to contribute to the lead before adding another lag.

Option 28.0 – Analog Input decimals (0-2) LV3

(Option 28.x is available on Firmware Version 1.204 and above (EK56))

Set the number of decimal places for analog input 1 on PA5-PA11.

Option 28.1 – Analog Input 1 zero (0-999) LV3

Set the number to be displayed when the input 1 current is 4mA.

Option 28.2 – Analog Input 1 span (0-999) LV3

Set the number to be displayed when the input 1 current is 20mA. Setting this parameter to a non-zero value also enables this item.

NOTE: Setting the span value enables the item.

On the 2-line display the values can then be seen by scrolling with the <> keys in RUN mode. They will always show up as “Analog Input 1” or “Analog Input 2” or “Analog Input 3”

On the touchscreen, the analog inputs appear on the right hand side list and can be added to the trends chart, etc. Also the touchscreen gives us the flexibility to type in text to name these inputs like the profiles can be (see screen configuration page on the touchscreen).

Option 28.3 – Analog Input 2 decimals (0-2) LV3

Set the number of decimal places for analog input 2 on PA6-PA11.

Option 28.4 – Analog Input 2 zero (0-999) LV3

Set the number to be displayed when the input 2 current is 4mA.

Option 28.5 – Analog Input 2 span (0-999) LV3

Set the number to be displayed when the input 2 current is 20mA. Setting this parameter to a non-zero value also enables this item. See NOTE under Option 28.2.

Option 28.6 – Analog Input 3 decimals (0-2) LV3

Set the number of decimal places for analog input 3 on PA7-PA11.



Option 28.7 – Analog Input 3 zero (0-999) LV3

Set the number to be displayed when the input 3 current is 4mA.

Option 28.8 – Analog Input 3 span (0-999) LV3

Set the number to be displayed when the input 3 current is 20mA. Setting this parameter to a non-zero value also enables this item. See NOTE under Option 28.2.

Option 29.0 – 4-20mA Output 1 function if not VSD LV3

Set this option to a value from the table below to enable an analog output to re-transmit system variables as a 4-20mA signal.

For example, if this output is to represent Flue Gas Temperature, then set a value of 22.

If this output has been assigned for VSD control in option 3.x, then this setting will be ignored.

Option 29.0, 29.3 and 29.6 function list:					
Value	Function		Value	Function	
0	Drive 0 Position		20	Spare	
1	Drive 1 Position		21	Boiler operating Setpoint	
2	Drive 2 Position		22	Flue gas temperature	
3	Drive 3 Position		23	Spare	
4	Drive 4 Position		24	Spare	
5	Drive 5 Position		25	Spare	
6	Drive 6 Position		26	Spare	
7	Drive 7 Position		27	Spare	
8	Drive 8 Position		28	Spare	
9	Drive 9 Position		29	Spare	
10	Spare		30	Burner modulation rate	
11	Spare		31	Spare	
12	Measured value		32	Spare	
13	Boiler Efficiency		33	Spare	
14	Inlet air temperature		34	Spare	
15	Flue gas oxygen level		35	Gas pressure	
16	Calculated flue gas CO2 level		36	Spare	
17	Spare		37	Flame signal	
18	Spare		38	Spare	
19	Spare		39	Spare	

Option 29.1 – Output1 zero value (4mA output) LV3

Set this option to the value which a 4mA output represents. Often this will be zero, but non zero values may be used. For example, a value of 100 may be better suited for a minimum Flue Gas Temperature to ensure best resolution.

If this output has been assigned for VSD control in option 3.x, then this setting will be ignored.

Option 29.2 – Output1 span value (20mA output) LV3

Set this option to the value which a 20mA output will represent. For example, value of 200 may be better suited for a maximum Flue Gas Temperature to ensure best resolution.

If this output has been assigned for VSD control in option 3.x, then this setting will be ignored.

Option 29.3 – 4-20mA Output 2 function if not VSD LV3

Set this option to a value from the table in 29.0 to enable an analog output to re-transmit system variables as a 4-20mA signal.

If this output has been assigned for VSD control in option 3.x, then this setting will be ignored.

Option 29.4 – Output2 zero value (4mA output) LV3

Set this option to the value which a 4mA output will represent. Often this will be zero, but non zero values may be used.

If this output has been assigned for VSD control in option 3.x, then this setting will be ignored.

Option 29.5 – Output2 span value (20mA output) LV3

Set this option to the value which a 20mA output will represent.

If this output has been assigned for VSD control in option 3.x, then this setting will be ignored.

Option 29.6 – 4-20mA Output 3 function if not VSD LV3

Set this option to a value from the table in 29.0 to enable an analog output to re-transmit system variables as a 4-20mA signal.

Option 29.7 – Output3 zero value (4mA output) LV3

Set this option to the value which a 4mA output will represent. Often this will be zero, but non zero values may be used.

Option 29.8 – Output3 span value (20mA output) LV3

Set this option to the value which a 20mA output will represent.

Commissioning Oxygen Trim**CAUTION**

- Use extreme care when entering flow values and oxygen setpoints. Incorrect setting of either could cause a hazardous situation to occur.
- If the flow values and oxygen setpoints have been up-loaded it is essential that the combustion is verified at each firing position to ensure a hazardous condition cannot occur.
- If using automatic oxygen trim commissioning it is the sole responsibility of the commissioning engineer to ensure the values calculated by the system are correct and will not cause a hazardous condition to occur.
- Once the oxygen trim is commissioned the trim flow limit must be set to ensure a fault in the oxygen trim will not cause a hazardous condition to occur.

In order for the oxygen trim system to function correctly, the following information must be entered:

1. Enter the appropriate option parameters from 30.0 to 42.6 as required.

With the burner off, enter commission mode (See section 5). To get the trim operational it is only necessary to address all option parameters with the **O 2**  symbol

Refer to options 30.0 thru 42.6 inclusive for details of parameters that may be entered.

2. Flow values and O2 setpoints. The flow and O2 values may be entered manually by the engineer (option 34.5), or the system can automatically calculate the values and enter them itself by using automatic trim commissioning (option 30.9).

If the automatic trim commissioning is performed, the engineer **MUST CHECK THAT THE VALUES ENTERED ARE VALID AND SAFE AFTER COMPLETION**. Additionally, the oxygen probe calibration should be checked before and after the procedure since the results depend highly upon measurements taken using the probe.

Before proceeding to enter the oxygen setpoints for each profile, a few points relating to the oxygen trim option should be considered.

- ♦ No trim is applied which would require a drive to move above its high fire position or below its low fire position.
- ♦ The flow for each profile position must be entered to ensure correct operation. The flow can either be measured for each profile position or calculated as a percentage with respect to the flow at high fire.

Using adjust ratio mode to modify and existing oxygen profile manually, follow steps 1 -12 listed below.



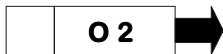
CAUTION

When using the adjust ratio mode, it is not possible for the unit to check drive positions at all times. It is the responsibility of the engineer to check that motors and valves are responding correctly

To adjust an oxygen setpoint or enter an air flow value in the firing range, use the following procedure. The manual calculation of the values to be entered here is covered in the next section.

1. Enter adjust ratio mode (See section 5.6).
2. Ensure oxygen trim is set to monitor only 30.5 = 0.
3. Use the scroll keys to select the desired setpoint and move up and/or down (A4 for example). The system will modulate to the selected setpoint and the number displayed will flash to indicate that the drives are modulating.
4. Wait for the number in the display to be illuminated steadily. This means that the drives have arrived at the selected setpoint.
5. The display will allow scrolling sideways to show flow and 'O2' in addition to the servo-motor positions to indicate that the system is expecting a flow value and an oxygen setpoint to be entered for the current profile setpoint. **To check the current oxygen measured value use the engineer's key EK44.**
6. Once the oxygen value is shown on the display use the scroll keys to position the '>' pointing to the 'O2' value use the UP/DOWN keys to adjust the oxygen setpoint as necessary. The display will start flashing to indicate that a change has been made.

7. If the new oxygen setpoint value is wanted, press the ENTER key. The value will be stored in memory.
8. If the new oxygen setpoint value is not wanted or an adjustment is not required, press the **NEXT** key to return to the value stored in memory.
9. Once the flow value is shown on the display use the scroll keys to position the '>' pointing to the 'FLO' value use the UP/DOWN keys to adjust the flow value as necessary. The display will show the air flow value, represented as a percentage of the air flow when the drives are at the **HIGH FIRE** position. The display will start flashing to indicate that a change has been made.
10. If the new flow value is not wanted or an adjustment is not required, press the **NEXT** key to return to the value stored in memory.
11. If it is necessary to adjust another setpoint in the firing range, repeat the above procedure from step 3.
12. After modifying an existing O2 trim profile, use Option 30.8 to reset the feed forward function. See option 30.8 for details.



Option 30.0 – Oxygen probe interface serial number LV3

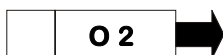
Each CANbus device connected to the system has a unique serial number allocated to it during production testing. This number cannot be changed and is used to uniquely identify each device.

When the system is in commissioning mode the serial number of all the connected devices is read by the PPC6000 and displayed as a 'list' to allow the selection of the relevant serial number unit to be made.

Scrolling the list in an upward direction will scroll through all oxygen probe interface options available until '0, nothing' is displayed. Enter this value to disconnect the specified device from any hardware.

Scrolling the list in a downward direction will stop when the currently selected item is displayed, and the value will stop flashing.

The display will show type information for each device present (such as 'O2 probe'), but the commissioning engineer should be aware of which serial number device is being used for which function.



Option 30.1 - Oxygen probe calibration offset value (0 – 999) LV3

This option parameter is only available if option 30.0 is non-zero.

This option parameter must be set to the calibration offset value that is specified with the Fireye probe supplied.

When using another manufacturer's O2 probe with a 4-20mA signal for O2 value, enter the O2 value for 4.0mA in this option parameter. Example 0% O2 = 000.



O 2 → **Option 30.2 - Oxygen probe calibration gain value (0 – 999) LV3**

This option parameter is only available if option 30.0 is non-zero.

This option parameter must be set to the calibration gain value that is specified with the Fireeye probe supplied.

When using another manufacturer's O₂ probe with a 4-20mA signal for the O₂ value, enter the O₂ value for 20.0mA in this option parameter. Example 21% O₂ = 210.

NOTE: If the value of 30.1 and 30.2 are both below 300, the PPC6000 will utilize the analog input on the NXO2INT terminals PL6 (+), PL7(-) to determine the O₂ signal.

Option 30.3 - Oxygen probe calibration gas concentration (0.00 – 9.99%) LV3

This option parameter is only available if option 30.0 is non-zero.

This option parameter must be set to the percentage oxygen concentration of the calibration gas used if the probe is to be calibrated using a reference gas (see option parameter 30.6). The range of this parameter is 0.00 - 9.99%.

O 2 → **Option 30.4 – Flue and inlet sensor temperature units (0 – 1) LV3**

This option parameter is only available if option 30.0 is non zero.

This option parameter must be set to select the units for display of both temperatures on either °C or °F.

Option parameter 30.4 value	Meaning
0	°C. The temperature value will be displayed in °C.
1	°F. The temperature value will be displayed in °F.

Please note that temperature limits **MUST** be set in the correct units.

If a 4-20mA input is used for O₂ signal, a NXIATS is required to calculate efficiency in conjunction with a K type thermocouple measuring flue temperature, connected to the appropriate inputs in the NXO2INT.

O 2 → **Option 30.5 - Oxygen input function (0/1) LV3**

This option parameter can only be set above 1 if option 30.0 is non zero.

Using this option parameter, it is possible to configure the oxygen input for a monitoring function or trim function. The two functions are explained below:

Option parameter 30.5 value	Meaning
0	Monitor only. The oxygen value is only used to provide a display of the measured oxygen level.
1	Closed loop trim. The oxygen value is used to provide both a display of the measured oxygen level and a feedback signal for closed loop trim control function of the PPC6000 series control.

Option 30.6 - Oxygen probe calibrate enable (0 – 2) LV3